minimum line length l that would result in an input impedance Z_{in}^{sc} equivalent to that of a capacitor of capacitance C_{eq} is

$$l = \frac{1}{\beta}\left[\pi - \tan^{-1}\left(\frac{1}{\omega C_{eq} Z_0}\right)\right] \quad \text{(m)}. \tag{2.91}$$

► These results imply that, through proper choice of the length of a short-circuited line, we can make them into equivalent capacitors and inductors of any desired reactance. ◄

Such a practice is indeed common in the design of microwave circuits and high-speed integrated circuits, because making an actual capacitor or inductor often is much more difficult than fabricating a microstrip transmission line with a shorted end on a circuit board.

Example 2-8: Equivalent Reactive Elements

Choose the length of a shorted 50 Ω lossless transmission line (**Fig. 2-20**) such that its input impedance at 2.25 GHz is identical to that of a capacitor with capacitance $C_{eq} = 4$ pF. The wave velocity on the line is $0.75c$.

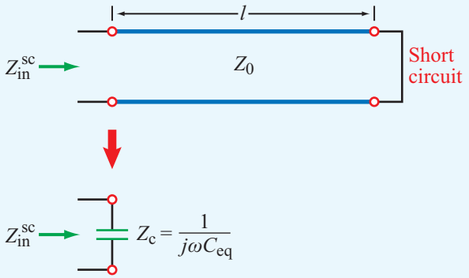


Figure 2-20 Shorted line as an equivalent capacitor (Example 2-8).

Solution: We are given

$$u_p = 0.75c = 0.75 \times 3 \times 10^8 = 2.25 \times 10^8 \text{ m/s},$$
$$Z_0 = 50\ \Omega,$$
$$f = 2.25 \text{ GHz} = 2.25 \times 10^9 \text{ Hz},$$
$$C_{eq} = 4 \text{ pF} = 4 \times 10^{-12} \text{ F}.$$

The phase constant is

$$\beta = \frac{2\pi}{\lambda} = \frac{2\pi f}{u_p} = \frac{2\pi \times 2.25 \times 10^9}{2.25 \times 10^8} = 62.8 \quad \text{(rad/m)}.$$

From Eq. (2.89), it follows that

$$\tan\beta l = -\frac{1}{Z_0 \omega C_{eq}} = -\frac{1}{50 \times 2\pi \times 2.25 \times 10^9 \times 4 \times 10^{-12}} = -0.354.$$

The tangent function is negative when its argument is in the second or fourth quadrants. The solution for the second quadrant is

$$\beta l_1 = 2.8 \text{ rad} \quad \text{or} \quad l_1 = \frac{2.8}{\beta} = \frac{2.8}{62.8} = 4.46 \text{ cm},$$

and the solution for the fourth quadrant is

$$\beta l_2 = 5.94 \text{ rad} \quad \text{or} \quad l_2 = \frac{5.94}{62.8} = 9.46 \text{ cm}.$$

We also could have obtained the value of l_1 by applying Eq. (2.91). The length l_2 is greater than l_1 by exactly $\lambda/2$. In fact, any length $l = 4.46 \text{ cm} + n\lambda/2$, where n is a positive integer, also is a solution.

2-8.2 Open-Circuited Line

With $Z_L = \infty$, as illustrated in **Fig. 2-21(a)**, we have $\Gamma = 1$ and $S = \infty$, and the voltage, current, and input impedance are given by

$$\tilde{V}_{oc}(d) = V_0^+[e^{j\beta d} + e^{-j\beta d}] = 2V_0^+ \cos\beta d, \tag{2.92a}$$
$$\tilde{I}_{oc}(d) = \frac{V_0^+}{Z_0}[e^{j\beta d} - e^{-j\beta d}] = \frac{2jV_0^+}{Z_0} \sin\beta d, \tag{2.92b}$$
$$Z_{in}^{oc} = \frac{\tilde{V}_{oc}(l)}{\tilde{I}_{oc}(l)} = -jZ_0 \cot\beta l. \tag{2.93}$$

Plots of these quantities are displayed in **Fig. 2-21** as a function of d for the voltage and current and as a function of l for the input impedance.

2-8.3 Application of Short-Circuit/Open-Circuit Technique

A network analyzer is a radio-frequency (RF) instrument capable of measuring the impedance of any load connected

导致输入阻抗 Z_{in}^{sc} 相当于电容 C_{eq} 的电容器的最小线路长度 l 为

$$l = \frac{1}{\beta}\left[\pi - \tan^{-1}\left(\frac{1}{\omega C_{eq} Z_0}\right)\right] \quad \text{(m)。} \tag{2.91}$$

► 这些结果意味着，通过正确选择短路线的长度，我们可以将它们制成任何所需电抗的等效电容器和电感器。 ◄

这种做法在微波电路和高速集成电路的设计中确实很常见，因为制造实际的电容器或电感器通常比在电路板上制造具有短路端的微带传输线困难得多。

示例 2-8：等效反应元素

选择短路的 50 Ω 无损传输线（图 2-20）的长度，使其在 2.25 GHz 时的输入阻抗与电容 $C_{eq} = 4$ pF 的电容器的输入阻抗相同。线上的波速为0.75c。

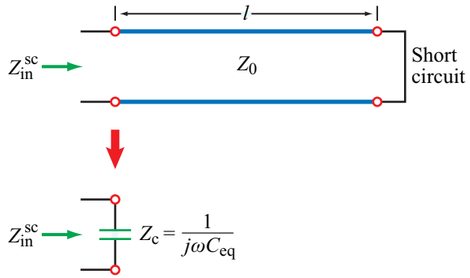


图 2-20 作为等效电容器的短路线（示例 2-8）。

解：我们给定

$$u_p = 0.75c = 0.75 \times 3 \times 10^8 = 2.25 \times 10^8 \text{ m/s},$$
$$Z_0 = 50\ \Omega,$$
$$f = 2.25 \text{ GHz} = 2.25 \times 10^9 \text{ Hz},$$
$$C_{eq} = 4 \text{ pF} = 4 \times 10^{-12} \text{ F}.$$

相位常数为

$$\beta = \frac{2\pi}{\lambda} = \frac{2\pi f}{u_p} = \frac{2\pi \times 2.25 \times 10^9}{2.25 \times 10^8} = 62.8 \quad \text{(弧度/米)。}$$

从方程。 (2.89)，由此可知

$$\tan\beta l = -\frac{1}{Z_0 \omega C_{eq}} = -\frac{1}{50 \times 2\pi \times 2.25 \times 10^9 \times 4 \times 10^{-12}} = -0.354.$$

当参数位于第二或第四象限时，正切函数为负。第二象限的解为

$$\beta l_1 = 2.8 \text{ rad} \quad \text{or} \quad l_1 = \frac{2.8}{\beta} = \frac{2.8}{62.8} = 4.46 \text{ cm},$$

第四象限的解为

$$\beta l_2 = 5.94 \text{ rad} \quad \text{or} \quad l_2 = \frac{5.94}{62.8} = 9.46 \text{ cm}.$$

我们还可以通过应用等式获得 l_1 的值。 (2.91)。长度 l_2 比 l_1 正好大 $\lambda/2$。事实上，任意长度$l = 4.46 \text{ cm} + n\lambda/2$，其中n为正整数，也是一个解。

2-8.2 开路线路

对于 $Z_L = \infty$，如图 2-21(a) 所示，我们有 $\Gamma = 1$ 和 $S = \infty$，电压、电流和输入阻抗由下式给出

$$\tilde{V}_{oc}(d) = V_0^+[e^{j\beta d} + e^{-j\beta d}] = 2V_0^+ \cos\beta d, \tag{2.92a}$$
$$\tilde{I}_{oc}(d) = \frac{V_0^+}{Z_0}[e^{j\beta d} - e^{-j\beta d}] = \frac{2jV_0^+}{Z_0} \sin\beta d, \tag{2.92b}$$
$$Z_{in}^{oc} = \frac{\tilde{V}_{oc}(l)}{\tilde{I}_{oc}(l)} = -jZ_0 \cot\beta l. \tag{2.93}$$

这些量的图显示在图 2-21 中，作为电压和电流的 d 函数以及输入阻抗的 l 函数。

2-8.3 短路/开路技术的应用

网络分析仪是一种射频 (RF) 仪器，能够测量任何连接负载的阻抗

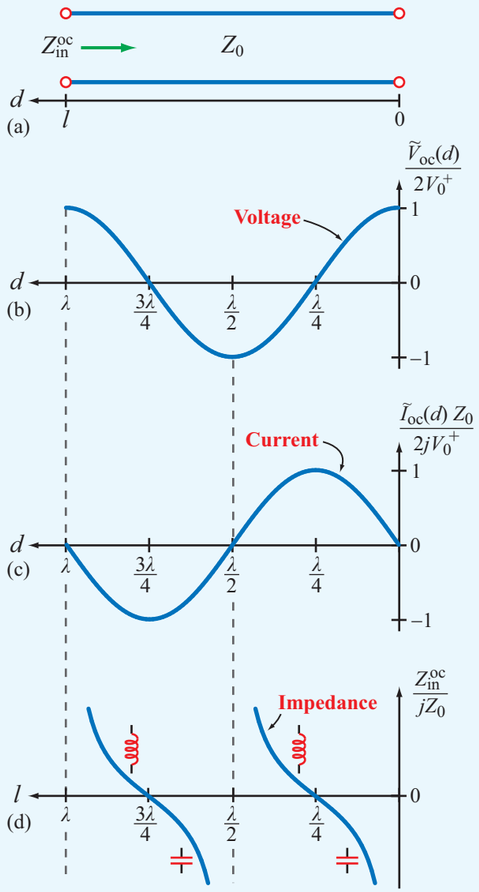


Figure 2-21 Transmission line terminated in an open circuit: (a) schematic representation, (b) normalized voltage on the line, (c) normalized current, and (d) normalized input impedance.

to its input terminal. When used to measure (1) Z_{in}^{sc}, which is the input impedance of a lossless line terminated in a short circuit, and (2) Z_{in}^{oc}, which is the input impedance of the line when terminated in an open circuit, the combination of the two measurements can be used to determine the characteristic impedance of the line Z_0 and its phase constant β. Indeed, the product of the expressions given by Eqs. (2.84) and (2.93) gives

$$Z_0 = \sqrt[+]{Z_{in}^{sc}\, Z_{in}^{oc}}, \tag{2.94}$$

and the ratio of the same expressions leads to

$$\tan\beta l = \sqrt{\frac{-Z_{in}^{sc}}{Z_{in}^{oc}}}. \tag{2.95}$$

Because of the π phase ambiguity associated with the tangent function, the length l should be less than or equal to $\lambda/2$ to provide an unambiguous result.

Example 2-9: Measuring Z_0 and β

Find Z_0 and β of a 57-cm long lossless transmission line whose input impedance was measured as $Z_{in}^{sc} = j40.42\ \Omega$ when terminated in a short circuit and as $Z_{in}^{oc} = -j121.24\ \Omega$ when terminated in an open circuit. From other measurements, we know that the line is between 3 and 3.25 wavelengths long.

Solution: From Eqs. (2.94) and (2.95),

$$Z_0 = \sqrt[+]{Z_{in}^{sc}\, Z_{in}^{oc}} = \sqrt{(j40.42)(-j121.24)} = 70\ \Omega,$$

$$\tan\beta l = \sqrt{\frac{-Z_{in}^{sc}}{Z_{in}^{oc}}} = \sqrt{\frac{1}{3}}.$$

Since l is between 3λ and 3.25λ, $\beta l = (2\pi l/\lambda)$ is between 6π radians and $(13\pi/2)$ radians. This places βl in the first quadrant (0 to $\pi/2$) radians. Hence, the only acceptable solution for $\tan\beta\ell = \sqrt{1/3}$ is $\beta l = \pi/6$ radians. This value, however, does not include the 2π multiples associated with the integer λ multiples of l. Hence, the true value of βl is

$$\beta l = 6\pi + \frac{\pi}{6} = 19.4 \qquad \text{(rad)},$$

in which case

$$\beta = \frac{19.4}{0.57} = 34 \qquad \text{(rad/m)}.$$

2-8.4 Lines of Length $l = n\lambda/2$

If $l = n\lambda/2$, where n is an integer,

$$\tan\beta l = \tan[(2\pi/\lambda)(n\lambda/2)] = \tan n\pi = 0.$$

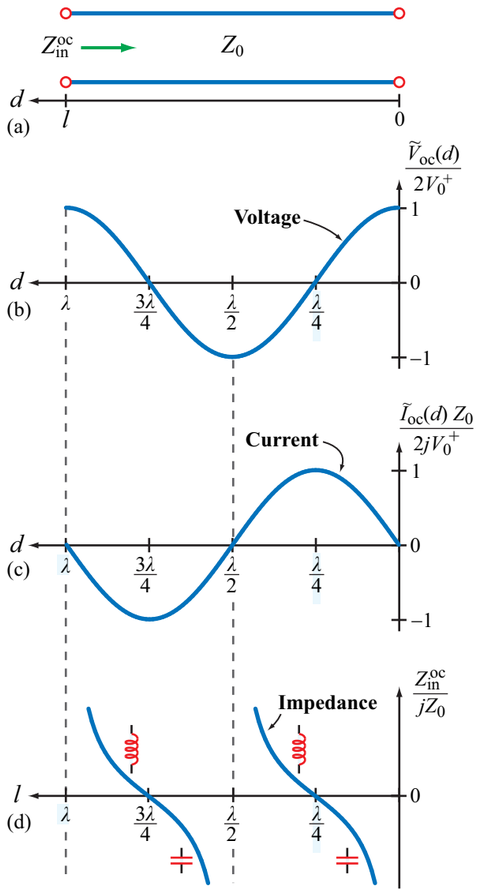


图 2-21 以开路端接的传输线：(a) 示意图，(b) 线路上的标准化电压，(c) 标准化电流，以及 (d) 标准化输入阻抗。

到其输入端子。当用于测量 (1) Z_{in}^{sc}（以短路方式端接的无损线路的输入阻抗）和 (2) Z_{in}^{oc}（以开路方式端接时线路的输入阻抗）时，两个测量值的组合可用于确定线路 Z_0 的特性阻抗及其相位常数 β。事实上，方程给出的表达式的乘积。 (2.84) 和 (2.93)

给出

$$Z_0 = \sqrt[+]{Z_{in}^{sc}\, Z_{in}^{oc}}, \tag{2.94}$$

并且相同表达式的比率导致

$$\tan\beta l = \sqrt{\frac{-Z_{in}^{sc}}{Z_{in}^{oc}}}. \tag{2.95}$$

由于与正切函数相关的 π 相位模糊性，长度 l 应小于或等于 λ/2 以提供明确的结果。

例 2-9：测量 Z_0 和 β

求 57 厘米长的无损传输线的 Z_0 和 β，其输入阻抗在短路端接时测量为 $Z_{in}^{sc} = j40.42\ \Omega$，在开路端接时测量为 $Z_{in}^{oc} = -j121.24\ \Omega$。从其他测量中，我们知道该线的长度在 3 到 3.25 个波长之间。

解决方案：从方程。 (2.94) 和 (2.95)，

$$Z_0 = \sqrt[+]{Z_{in}^{sc}\, Z_{in}^{oc}} = \sqrt{(j40.42)(-j121.24)} = 70\ \Omega,$$

$$\tan\beta l = \sqrt{\frac{-Z_{in}^{sc}}{Z_{in}^{oc}}} = \sqrt{\frac{1}{3}}.$$

由于 l 介于 3λ 和 3.25λ 之间，因此 $\beta l = (2\pi l/\lambda)$ 介于 6π 弧度和 $(13\pi/2)$ 弧度之间。这会将 βl 置于第一象限（0 到 π/2）弧度内。因此，$\tan\beta\ell = \sqrt{1/3}$ 唯一可接受的解决方案是 $\beta l = \pi/6$ 弧度。然而，该值不包括与 l 的整数 λ 倍数相关的 2π 倍数。因此，βl 的真实值为

$$\beta l = 6\pi + \frac{\pi}{6} = 19.4 \qquad \text{(rad)},$$

在这种情况下

$$\beta = \frac{19.4}{0.57} = 34 \qquad \text{(rad/m)}.$$

2-8.4 长度为 $l = n\lambda/2$ 的线

如果 $l = n\lambda/2$，其中 n 是整数，

$$\tan\beta l = \tan[(2\pi/\lambda)(n\lambda/2)] = \tan n\pi = 0.$$

Consequently, Eq. (2.79) reduces to

$$Z_{in} = Z_L, \qquad \text{for } l = n\lambda/2, \tag{2.96}$$

which means that a half-wavelength line (or any integer multiple of $\lambda/2$) does not modify the load impedance.

2-8.5 Quarter-Wavelength Transformer

Another case of interest is when the length of the line is a quarter-wavelength (or $\lambda/4 + n\lambda/2$, where $n = 0$ or a positive integer), corresponding to $\beta l = (2\pi/\lambda)(\lambda/4) = \pi/2$. From Eq. (2.79), the input impedance becomes

$$Z_{in} = \frac{Z_0^2}{Z_L}, \qquad \text{for } l = \lambda/4 + n\lambda/2. \tag{2.97}$$

The utility of such a quarter-wave transformer is illustrated by Example 2-10.

Example 2-10: $\lambda/4$ Transformer

A 50 Ω lossless transmission line is to be matched to a resistive load impedance with $Z_L = 100$ Ω via a quarter-wave section as shown in **Fig. 2-22**, thereby eliminating reflections along the feedline. Find the required characteristic impedance of the quarter-wave transformer.

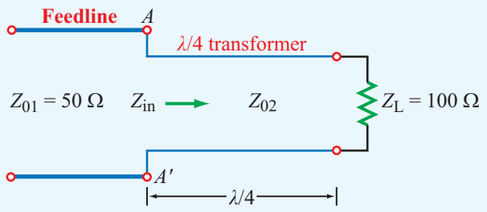


Figure 2-22 Configuration for Example 2-10.

Solution: To eliminate reflections at terminal AA', the input impedance Z_{in} looking into the quarter-wave line should be equal to Z_{01}, which is the characteristic impedance of the feedline. Thus, $Z_{in} = 50$ Ω. From Eq. (2.97),

$$Z_{in} = \frac{Z_{02}^2}{Z_L},$$

or

$$Z_{02} = \sqrt{Z_{in} Z_L} = \sqrt{50 \times 100} = 70.7\ \Omega.$$

Whereas this eliminates reflections on the feedline, it does not eliminate them on the $\lambda/4$ line. However, since the lines are lossless, all of the power incident on AA' will end up getting transferred into the load Z_L.

► In this example, Z_L is purely resistive. To apply the $\lambda/4$ transformer technique to match a transmission line to a load with a complex impedance, a more elaborate procedure is required (Section 2-11). ◄

2-8.6 Matched Transmission Line: $Z_L = Z_0$

For a matched lossless transmission line with $Z_L = Z_0$, (1) the input impedance $Z_{in} = Z_0$ for all locations d on the line, (2) $\Gamma = 0$, and (3) all of the incident power is delivered to the load, regardless of the line length l. A summary of the properties of standing waves is given in **Table 2-4**.

Concept Question 2-10: What is the difference between the characteristic impedance Z_0 and the input impedance Z_{in}? When are they the same?

Concept Question 2-11: What is a quarter-wave transformer? How can it be used?

Concept Question 2-12: A lossless transmission line of length l is terminated in a short circuit. If $l < \lambda/4$, is the input impedance inductive or capacitive?

Concept Question 2-13: What is the input impedance of an infinitely long line?

Concept Question 2-14: If the input impedance of a lossless line is inductive when terminated in a short circuit, will it be inductive or capacitive when the line is terminated in an open circuit?

Exercise 2-14: A 50 Ω lossless transmission line uses an insulating material with $\varepsilon_r = 2.25$. When terminated in an open circuit, how long should the line be for its input impedance to be equivalent to a 10 pF capacitor at 50 MHz?

Answer: $l = 9.92$ cm. (See EM.)

因此，等式。(2.79) 减少为

$$Z_{in} = Z_L, \qquad \text{for } l = n\lambda/2, \tag{2.96}$$

这意味着半波长线（或 λ/2 的任何整数倍）不会改变负载阻抗。

2-8.5 四分之一波长变压器

另一种有趣的情况是当线的长度为四分之一波长（或 λ/4 + nλ/2，其中 n = 0 或正整数）时，对应于 βl = (2π/λ)(λ/4) = π/2。从方程。(2.79)，输入阻抗变为

$$Z_{in} = \frac{Z_0^2}{Z_L}, \qquad \text{for } l = \lambda/4 + n\lambda/2. \tag{2.97}$$

例 2-10 说明了这种四分之一波长变压器的实用性。

例 2-10：λ/4 变压器

50 Ω 无损传输线将通过四分之一波长部分与 $Z_L = 100$ Ω 的电阻负载阻抗匹配，如图 2-22 所示，从而消除沿馈线的反射。找到四分之一波长变压器所需的特性阻抗。

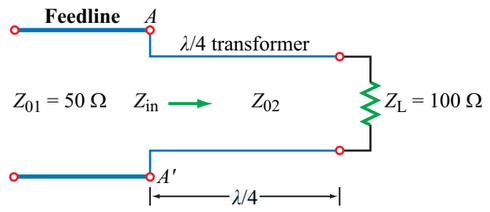


图 2-22 示例 2-10 的配置。

解决方案：为了消除端子 AA′ 处的反射，四分之一波长线路的输入阻抗 Z_{in} 应等于 Z_{01}，即馈线的特性阻抗。因此，$Z_{in} = 50\Omega$。从方程。(2.97)，

$$Z_{in} = \frac{Z_{02}^2}{Z_L},$$

或者

$$Z_{02} = \sqrt{Z_{in} Z_L} = \sqrt{50 \times 100} = 70.7\ \Omega.$$

虽然这消除了馈线上的反射，但它并没有消除 λ/4 线上的反射。然而，由于线路是无损的，所有入射到 AA′ 上的功率最终都会转移到负载 Z_L 中。

► 在此示例中，Z_L 是纯电阻性的。要应用 λ/4 变压器技术将传输线与具有复阻抗的负载相匹配，需要更复杂的过程（第 2-11 节）。◄

2-8.6 匹配传输线：$Z_L = Z_0$

对于具有 $Z_L = Z_0$ 的匹配无损传输线，(1) 线路上所有位置 d 的输入阻抗 $Z_{in} = Z_0$，(2) $\Gamma = 0$，以及 (3) 无论线路长度 l，所有入射功率都传递到负载。表 2-4 总结了驻波的特性。

概念问题2-10：特性阻抗Z_0和输入阻抗Z_{in}有什么区别？他们什么时候一样了？

概念问题 2-11：什么是四分之一波变压器？如何使用？

概念问题 2-12：长度为 l 的无损传输线以短路终止。如果$l < \lambda/4$，输入阻抗是感性的还是容性的？

概念问题 2-13：无限长线路的输入阻抗是多少？

概念问题 2-14：如果无损线路的输入阻抗在短路端接时是感性的，那么当线路在开路端接时它是感性的还是容性的？

练习 2-14：50 Ω 无损传输线使用 $\varepsilon_r = 2.25$ 的绝缘材料。当以开路端接时，线路应有多长才能使其输入阻抗相当于 50 MHz 下的 10 pF 电容器？

答案：l = 9.92 厘米。（参见EM。）

Table 2-4 Properties of standing waves on a lossless transmission line.

Voltage maximum **Voltage minimum**	$\|\tilde{V}\|_{\max} = \|V_0^+\|[1+\|\Gamma\|]$ $\|\tilde{V}\|_{\min} = \|V_0^+\|[1-\|\Gamma\|]$
Positions of voltage maxima (also positions of current minima)	$d_{\max} = \dfrac{\theta_r\lambda}{4\pi} + \dfrac{n\lambda}{2}, \quad n = 0, 1, 2, \ldots$
Position of first maximum (also position of first current minimum)	$d_{\max} = \begin{cases} \dfrac{\theta_r\lambda}{4\pi}, & \text{if } 0 \le \theta_r \le \pi \\ \dfrac{\theta_r\lambda}{4\pi} + \dfrac{\lambda}{2}, & \text{if } -\pi \le \theta_r \le 0 \end{cases}$
Positions of voltage minima (also positions of current maxima)	$d_{\min} = \dfrac{\theta_r\lambda}{4\pi} + \dfrac{(2n+1)\lambda}{4}, \quad n = 0, 1, 2, \ldots$
Position of first minimum (also position of first current maximum)	$d_{\min} = \dfrac{\lambda}{4}\left(1 + \dfrac{\theta_r}{\pi}\right)$
Input impedance	$Z_{in} = Z_0\left(\dfrac{z_L + j\tan\beta l}{1 + jz_L\tan\beta l}\right) = Z_0\left(\dfrac{1+\Gamma_l}{1-\Gamma_l}\right)$
Positions at which Z_{in} is real	at voltage maxima and minima
Z_{in} at voltage maxima	$Z_{in} = Z_0\left(\dfrac{1+\|\Gamma\|}{1-\|\Gamma\|}\right)$
Z_{in} at voltage minima	$Z_{in} = Z_0\left(\dfrac{1-\|\Gamma\|}{1+\|\Gamma\|}\right)$
Z_{in} of short-circuited line	$Z_{in}^{sc} = jZ_0\tan\beta l$
Z_{in} of open-circuited line	$Z_{in}^{oc} = -jZ_0\cot\beta l$
Z_{in} of line of length $l = n\lambda/2$	$Z_{in} = Z_L, \quad n = 0, 1, 2, \ldots$
Z_{in} of line of length $l = \lambda/4 + n\lambda/2$	$Z_{in} = Z_0^2/Z_L, \quad n = 0, 1, 2, \ldots$
Z_{in} of matched line	$Z_{in} = Z_0$
$\|V_0^+\|$ = amplitude of incident wave; $\Gamma = \|\Gamma\|e^{j\theta_r}$ with $-\pi < \theta_r < \pi$; θ_r in radians; $\Gamma_l = \Gamma e^{-j2\beta l}$.	

Exercise 2-15: A 300 Ω feedline is to be connected to a 3-m long, 150 Ω line terminated in a 150 Ω resistor. Both lines are lossless and use air as the insulating material, and the operating frequency is 50 MHz. Determine (a) the input impedance of the 3-m long line, (b) the voltage standing-wave ratio on the feedline, and (c) the characteristic impedance of a quarter-wave transformer were it to be used between the two lines in order to achieve $S = 1$ on the feedline. (See EM.)

Answer: (a) $Z_{in} = 150\ \Omega$, (b) $S = 2$, (c) $Z_0 = 212.1\ \Omega$.

Exercise 2-16: Through multiple trials, it was determined that a load with unknown impedance Z_L can be perfectly matched to a feedline with $Z_{in} = 50\ \Omega$ by using a $\lambda/4$ transformer section with characteristic impedance of 60 Ω. What is Z_L?

Answer: $Z_L = 72\ \Omega$.

2-8 SPECIAL CASES OF THE LOSSLESS LINE 79

表2-4 无损传输线上的驻波特性。

Voltage maximum	$\|\tilde{V}\|_{\max} = \|V_0^+\|[1+\|\Gamma\|]$
Voltage minimum	$\|\tilde{V}\|_{\min} = \|V_0^+\|[1-\|\Gamma\|]$
Positions of voltage maxima (also positions of current minima)	$d_{\max} = \dfrac{\theta_r\lambda}{4\pi} + \dfrac{n\lambda}{2}, \quad n = 0, 1, 2, \ldots$
Position of first maximum (also position of first current minimum)	$d_{\max} = \begin{cases} \dfrac{\theta_r\lambda}{4\pi}, & \text{if } 0 \le \theta_r \le \pi \\ \dfrac{\theta_r\lambda}{4\pi} + \dfrac{\lambda}{2}, & \text{if } -\pi \le \theta_r \le 0 \end{cases}$
Positions of voltage minima (also positions of current maxima)	$d_{\min} = \dfrac{\theta_r\lambda}{4\pi} + \dfrac{(2n+1)\lambda}{4}, \quad n = 0, 1, 2, \ldots$
Position of first minimum (also position of first current maximum)	$d_{\min} = \dfrac{\lambda}{4}\left(1 + \dfrac{\theta_r}{\pi}\right)$
Input impedance	$Z_{in} = Z_0\left(\dfrac{z_L + j\tan\beta l}{1 + jz_L\tan\beta l}\right) = Z_0\left(\dfrac{1+\Gamma_l}{1-\Gamma_l}\right)$
Positions at which Z_{in} is real	at voltage maxima and minima
Z_{in} at voltage maxima	$Z_{in} = Z_0\left(\dfrac{1+\|\Gamma\|}{1-\|\Gamma\|}\right)$
Z_{in} at voltage minima	$Z_{in} = Z_0\left(\dfrac{1-\|\Gamma\|}{1+\|\Gamma\|}\right)$
Z_{in} of short-circuited line	$Z_{in}^{sc} = jZ_0\tan\beta l$
Z_{in} of open-circuited line	$Z_{in}^{oc} = -jZ_0\cot\beta l$
Z_{in} of line of length $l = n\lambda/2$	$Z_{in} = Z_L, \quad n = 0, 1, 2, \ldots$
Z_{in} of line of length $l = \lambda/4 + n\lambda/2$	$Z_{in} = Z_0^2/Z_L, \quad n = 0, 1, 2, \ldots$
Z_{in} of matched line	$Z_{in} = Z_0$
$\|V_0^+\|$ = amplitude of incident wave; $\Gamma = \|\Gamma\|e^{j\theta_r}$ with $-\pi < \theta_r < \pi$; θ_r in radians; $\Gamma_l = \Gamma e^{-j2\beta l}$.	

练习 2-15：一条 300 Ω 馈线将连接到一条 3 米长、150 Ω 线路，该线路端接一个 150 Ω 电阻器。两条线路都是无损的，并使用空气作为绝缘材料，工作频率为 50 MHz。确定 (a) 3 米长线路的输入阻抗，(b) 馈线上的电压驻波比，以及 (c) 在两条线路之间使用四分之一波长变压器的特性阻抗，以便在馈线上实现 S = 1。（参见EM。）

答案：(a) $Z_{in} = 150\ \Omega$，(b) S = 2，(c) $Z_0 = 212.1\ \Omega$。

练习 2-16：通过多次试验，确定阻抗 Z_L 未知的负载可以通过使用特性阻抗为 60 Ω 的 λ/4 变压器部分与 $Z_{in} = 50\ \Omega$ 的馈线完美匹配。Z_L是什么?

答案：$Z_L = 72\ \Omega$。

Technology Brief 3: Microwave Ovens

Percy Spencer, while working for Raytheon in the 1940s on the design and construction of ***magnetrons*** for radar, observed that a chocolate bar that had unintentionally been exposed to microwaves had melted in his pocket. The process of cooking by microwave was patented in 1946 and by the 1970s, microwave ovens had become standard household items.

Microwave Absorption

A microwave is an ***electromagnetic wave*** whose frequency lies in the 300 MHz–300 GHz range (see **Fig. 1-16**). When a material containing water is exposed to microwaves, the water molecule reacts by rotating itself in order to align its own ***electric dipole*** along the direction of the oscillating electric field of the microwave. The rapid vibration motion creates heat in the material, resulting in the conversion of microwave energy into thermal energy. The absorption coefficient of water, $\alpha(f)$, exhibits a microwave spectrum that depends on the temperature of the water and the concentration of dissolved salts and sugars present in it. If the frequency f is chosen such that $\alpha(f)$ is high, the water-containing material absorbs much of the microwave energy passing through it and converts it to heat. However, this also means that most of the energy is absorbed by a thin surface layer of the material with not much energy remaining to heat deeper layers. The penetration depth δ_p of a material, defined as $\delta_p = 1/2\alpha$, is a measure of how deep the power carried by an EM wave can penetrate into the material. Approximately 95% of the microwave energy incident upon a material is absorbed by the surface layer of thickness $3\delta_p$. **Figure TF3-1** displays calculated spectra of δ_p for pure water and two materials with different water contents.

▶ The frequency most commonly used in microwave ovens is 2.45 GHz. The magnitude of δ_s at 2.45 GHz varies beween ~2 cm for pure water and 8 cm for a material with a water content of only 20%. ◀

This is a practical range for cooking food in a microwave oven; at much lower frequencies, the food is not a good absorber of energy (in addition to the fact that the design

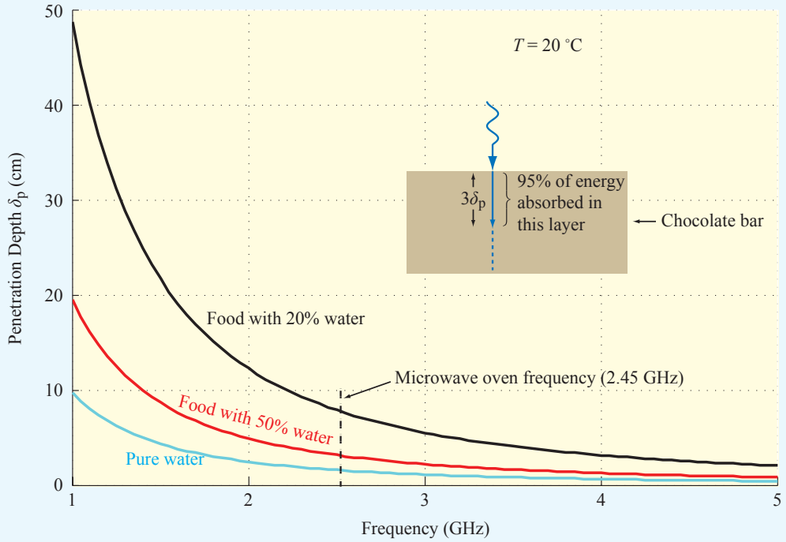


Figure TF3-1 Penetration depth as a function of frequency (1–5 GHz) for pure water and two foods with different water contents.

技术简介 3：微波炉

20 世纪 40 年代，珀西·斯宾塞 (Percy Spencer) 在雷神公司 (Raytheon) 从事雷达磁控管的设计和建造工作时，发现他口袋里的一块巧克力棒在无意中接触到微波后融化了。微波炉烹饪工艺于 1946 年获得专利，到 20 世纪 70 年代，微波炉已成为标准家庭用品。

微波吸收

微波是一种频率在 300 MHz 至 300 GHz 范围内的电磁波（见图 1-16）。当含水材料暴露于微波时，水分子通过自身旋转而发生反应，以便沿着微波振荡电场的方向排列其自身的电偶极子。快速振动运动在材料中产生热量，从而将微波能转化为热能。水的吸收系数 $\alpha(f)$ 呈现出取决于水温和溶解物浓度的微波光谱。

其中含有盐和糖。如果选择频率f使得$\alpha(f)$较高，则含水材料吸收大量穿过它的微波能量并将其转化为热量。然而，这也意味着大部分能量被材料的薄表面层吸收，没有太多能量剩余来加热更深的层。材料的穿透深度 δ_p 定义为 $\delta_p = 1/2\alpha$，是电磁波携带的功率可以穿透材料多深的量度。入射到材料上的微波能量大约 95% 被厚度为 $3\delta_p$ 的表面层吸收。图 TF3-1 显示了纯水和两种不同含水量材料的 δ_p 计算光谱。

▶ 微波炉最常用的频率是 2.45 GHz。 2.45 GHz 时，δ_s 的幅度在纯水的 ~ 2 cm 和含水量仅为 20% 的材料的 8 cm 之间变化。 ◀

这是一个实用的微波炉烹饪食物的系列；在低得多的频率下，食物并不是很好的能量吸收剂（此外，设计

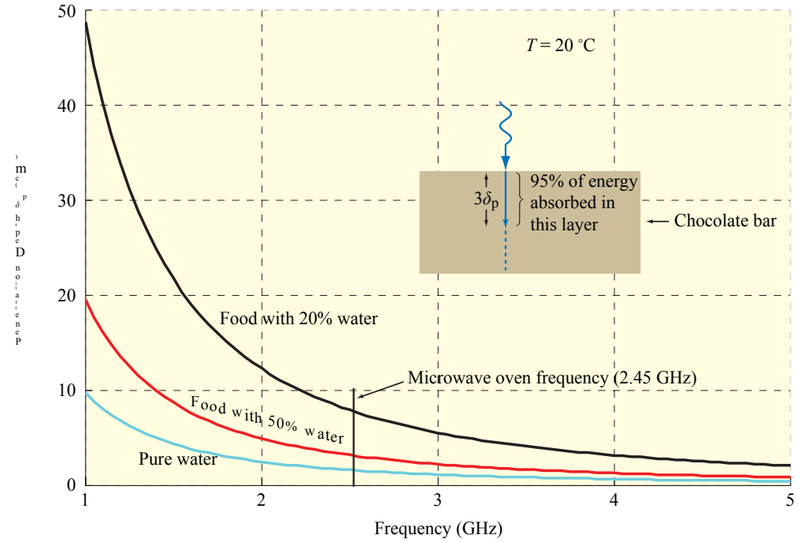


图 TF3-1 纯水和两种不同含水量食品的渗透深度与频率 (1–5 GHz) 的函数关系。